Draft of October 30, 2026 — not yet submitted.

Digital Object Identifier NN.NNNN/ACCESS.YYYY.DOI

VALDOMIRO DE OLIVEIRA SOUZA JR.[1] **and RAFAEL KUNST**[1]

[1]Applied Computing Graduate Program (PPGCA), Universidade do Vale do Rio dos Sinos (Unisinos), São Leopoldo 93022-750, Brazil (e-mail: valdomiro.souza@gmail.com, valdomiros@edu.unisinos.br; rafaelkunst@unisinos.br)

Corresponding author: Valdomiro de Oliveira Souza Jr. (e-mail: valdomiros@edu.unisinos.br).

This work was conducted within the Applied Computing Graduate Program (PPGCA) at Unisinos, Brazil. This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Incident response (IR) in IT operations is measured by Mean Time to Detect (MTTD) and Mean Time to Recovery (MTTR) and burdened by operator cognitive load. Agentic AI systems, which combine Large Language Models (LLMs) with planning, memory, tool use, and human oversight, are increasingly proposed as copilots for SRE, NOC, and SOC teams. This systematic literature review (SLR), following Kitchenham's guidelines and the PRISMA 2020 statement, asks whether the published evidence supports that promise. Across three update cycles over a foundational corpus, 126 candidate studies (2020–2026) were screened, 74 were assessed in full text, and **65** were included. The review is LLM-assisted but human-decided: three supervised LLM assistants (Claude, ChatGPT, Gemini) drafted each assessment independently (Fleiss' $\kappa = 0.60$; 66/74 unanimous), and every decision was made and reviewed by the authors. Four primary studies measure time metrics against executed baselines, reporting reductions in MTTD (82–83%), MTTR (63–72%), and time-to-containment (38%); two of them appear in bottom-quartile or unindexed venues, and the only production deployment is vendor-affiliated. Elsewhere, the evidence consists of detection proxies (accuracy, F1, AUC) and qualitative claims. Within the review's scope, no study measured operator cognitive load in the presence of an agentic copilot in IT incident response, although validated instruments exist in adjacent domains. The claim that agentic copilots reduce MTTD and MTTR is therefore supported by early primary evidence only. We outline an agenda of multi-site field evaluations that report nominal MTTD and MTTR against their own baselines together with validated workload measures.

Agentic AI, AIOps, cognitive load, incident response, LLM agents, multi-agent systems, MTTD, MTTR, SOC, systematic literature review.

AIOps	Artificial Intelligence for IT Operations
HITL	Human-in-the-Loop (human approval inside the action loop)
HOTL	Human-on-the-Loop (human supervision over an autonomous loop)
IR	Incident Response
LLM	Large Language Model
MAS	Multi-Agent System
MCP	Model Context Protocol
ML	Machine Learning
MTTD	Mean Time to Detect
MTTR	Mean Time to Recovery
NOC	Network Operations Center
QA	Quality Assessment (criteria QA1–QA4)
RAG	Retrieval-Augmented Generation
RCA	Root Cause Analysis
RL	Reinforcement Learning
SIEM	Security Information and Event Management
SLM	Small Language Model
SOAR	Security Orchestration, Automation, and Response
SOC(aaS)	Security Operations Center (as a Service)
SRE	Site Reliability Engineering / Engineer

MODERN IT operations teams face a structural asymmetry: distributed, cloud-native systems generate telemetry at machine speed, while detection, diagnosis, and remediation remain bounded by human cognition. The operational cost of this asymmetry is captured by two indicators — *Mean Time to Detect* (MTTD) and *Mean Time to Recovery* (MTTR) — and by a human cost that is harder to see on dashboards: alert fatigue and sustained cognitive load on SRE, NOC, and SOC personnel. In security operations, industry data place the average data-breach life cycle at 277 days (204 to identify, 73 to contain) [1], as relayed in the corpus [2]; this security-breach life cycle is related to, but distinct from, the availability-incident MTTR of SRE practice.

Agentic AI — AI systems that pursue goals with some degree of autonomy by combining LLM-based reasoning with planning, memory, tool use, and human oversight [3], [4] — has been proposed as a new operating model for incident response: not full automation, but a *copilot* that perceives, reasons, acts, and learns alongside the human operator. The promise is threefold: earlier detection (MTTD), faster remediation (MTTR), and a lighter day-to-day workload for operations teams.

This article asks whether the published evidence supports that promise. It reports a systematic literature review (SLR) conducted under the guidelines of Kitchenham and Charters [5], with study selection reported in the PRISMA 2020 format [6], scope delimited with the PICOC framework [7], and quality assessment mirroring the DARE criteria [8] — retained for comparability with the foundational review's assessments; primary-study-specific checklists such as Dybå and Dingsøyr's [9] are a planned refinement for future cycles. The review was executed as an *update review* [10] in three cycles over a foundational corpus of 19 studies, processing 126 candidate studies in total. The review is *LLM-assisted and human-decided*. Every full-text assessment was first drafted by three LLM-based research assistants from different model families (Claude, ChatGPT, Gemini), run independently under the same written protocol, and their mutual agreement was quantified with Cohen's [11] and Fleiss' [12] kappa as a consistency check. The assistants recommend; they do not decide: all screening, eligibility, and inclusion decisions, the adjudication of every divergence, and the review of every assessment were performed by the authors — the master's student conducting the review, under the supervision and review of the advisor.

The contributions are:

1) a consolidated corpus of 65 studies (2020–2026) on Agentic AI for incident response and adjacent IT operations, organized into a taxonomy of agentic paradigms (Section IV);
2) triple-verified bibliometrics (OpenAlex, Crossref, Scopus) and an internal cross-citation map of the corpus (65 citing edges across all three cycles) identifying its conceptual hubs;
3) evidence synthesis tracking the review's central claim across cycles: **no nominal MTTD/MTTR measurement with an own baseline existed through cycle 2; a first four-study cluster emerges in cycle 3, half of it below the review's venue bar** (Section IV-C), together with a catalog of the proxies otherwise used;
4) a bridge to the human-factors literature: a cluster of studies that measure operator cognitive load directly — yet never in combination with an agentic copilot in IT incident response (Section IV-D);
5) a methodological contribution: an LLM-assisted, human-decided assessment protocol — three supervised LLM assistants whose recommendations are adjudicated by the authors — in which the assistants' residual disagreements (8 of 74 assessments, all from the first two cycles) are traceable to three explicit protocol rules rather than to content interpretation — rules that, once ratified and rendered into the normative prompt text, produced a *divergence-free* third cycle (Section IV-F).

A. FROM AIOPS TO AGENTIC AI

AIOps applies machine learning to IT operations data — logs, metrics, traces — for anomaly detection, root cause analysis (RCA), and remediation. Surveys in the corpus trace its evolution from knowledge-based automation to ML pipelines [13] and, after 2023, to LLM-centered architectures [14]. The agentic turn adds goal pursuit under governance: taxonomies distinguish *AI agents* (tool-using, single-task) from *Agentic AI* (multi-step autonomy with planning and memory) [3], and reference surveys map its technologies, applications, and societal implications [4], [15], [16].

B. INCIDENT RESPONSE AND ITS METRICS

Incident response spans detection, triage, diagnosis (RCA), remediation, and post-incident learning. MTTD and MTTR are its canonical outcome metrics; alert fatigue and operator cognitive load are its canonical human costs. In security operations, the same cycle appears as SOC triage and SOAR playbooks [17], [2]. A persistent finding of this review — anticipated here — is that primary studies rarely instrument these nominal metrics, reporting detection-quality proxies instead.

C. THE FOUNDATIONAL CORPUS AND ITS HUBS

The present review updates a foundational SLR (19 studies, P01–P19; the protocol identifiers Pxx used throughout, notably in Figs. 1 and 3, are mapped to the bibliography in the Appendix, Table 7). Cross-citation analysis over the full corpus (Section IV-E) shows that three foundational surveys act as conceptual hubs — [4] and [18] (7 internal citations each) and [3] (6) — confirming their role as the definitional substrate on which the empirical studies build.

The protocol follows Kitchenham and Charters [5]: planning (PICOC, research questions, search string), conducting

PICOC delimitation of the review.

Element	Definition
Population	Complex software systems and the teams operating them (SRE, NOC, SOC); microservices; IT operations pipelines.
Intervention	Agentic AI as copilot or autonomous agent: LLM-based agents, multi-agent systems, LLM4AIOps, the perceive–reason–act–learn loop.
Comparison	Traditional automation, non-agentic AIOps, or human baselines — *declared only with an executed empirical baseline* (rule v1.1.0).
Outcomes	MTTD, MTTR, observability, detection accuracy, alert fatigue, and operator cognitive load (complementary outcome).
Context	The incident-response life cycle in IT operations and cybersecurity; HITL/HOTL supervision; production vs. simulation.

(search, screening, quality assessment, extraction), and reporting (this article). All protocol artifacts — prompts, extraction sheets, per-study assessments, and machine-readable result tables — are version-controlled in the research repository.

Division of roles. Large language models act in this review only as *supervised research assistants*: they assisted candidate identification (cycle 1), drafted the full-text PICOC extractions, and drafted the full-text assessments. None of them takes a decision. Every screening, eligibility, and inclusion decision, every protocol amendment and ratified rule, every adjudication of a divergence among the assistants, and the review of every LLM-drafted assessment and extraction were performed by the authors — the master's student as review conductor, with each decision reviewed and approved by the advisor.

A. PICOC

Table 1 shows the PICOC delimitation [7]. A protocol rule (v1.1.0), introduced after a calibration of the PICOC extraction itself across the three LLM assistants, requires that *Comparison* be recorded as declared **only when the study executes an empirical comparison against a concrete baseline**; purely conceptual contrasts (e.g., "Agentic AI vs. traditional AI" in prose) are recorded as not applicable. This rule eliminated 100% of the inter-assistant divergences observed in the Comparison field.

B. RESEARCH QUESTIONS

- **RQ1 (Context/Definitions).** Which capabilities and autonomy levels (planning, memory, tool use, human supervision) characterize Agentic AI, and how do they affect decision-making in complex environments?
- **RQ2 (Engineering/Architecture).** Which architectures and mechanisms (orchestration, memory, tools, guardrails, observability) sustain Agentic AI in production?
- **RQ3 (Evidence/Benefits).** Which quantitative and qualitative evidence reports benefits, and which metrics are used (response time, decision quality, operational load), especially in incident response?
- **RQ4 (Challenges/Ethics).** Which risks and challenges (security, robustness, governance, accountability) limit responsible use in critical systems?
- **RQ5 (Gaps).** Which gaps and future directions (evaluation/benchmarking, threat models, governance, observability, alignment) are identified?

C. SEARCH AND IDENTIFICATION

The search string (reporting guided by PRISMA-S [19]) was derived from the cross-study PICOC synthesis as two mandatory blocks — *Intervention* (agentic AI, LLM agents, multi-agent systems, AIOps copilots) AND *Context/Population* (incident response, IT operations, SRE/NOC/SOC, observability) — with *Outcomes* deliberately optional, since only 1 of 38 calibration studies named MTTD/MTTR in its metadata. The string was calibrated against the known corpus and validated by real execution on OpenAlex (≈49,700 records) and on the Scopus Search API (12,783 records, 2020+, English), achieving 13/14 recall of the included calibration set in both engines with the same single miss. Candidate identification combined assisted search (cycle 1: three LLM assistants supervised by the authors, ≈51 raw records), author-driven discovery (cycle 2: 49 candidates), and re-execution of the validated query in five database sweeps (cycle 3: 37 new candidates). Backward/forward snowballing [20] is mandatory in the protocol because one included study is not indexed by Scopus and one is not retrieved by the main query; the cycle-3 snowballing pass (backward over OpenAlex-deposited references plus a full-text scan of the entire PDF archive; forward over the citers of the one high-citation candidate excluded by the series-stratum rule) yielded no additional eligible candidate.

D. SELECTION CRITERIA

Inclusion required: publication year ≥2020 (I1); an identifiable peer-reviewed venue (I2); a venue-quality threshold (I3): the venue's **Scopus CiteScore percentile mapped to quartiles, with Q1–Q3 (≥25th percentile) eligible** and bottom-quartile or unindexed venues ineligible; at least one citation in any of three indexers, with a recency exception for papers younger than 12 months (I5); and thematic adherence — an *agentic* approach in the *IR/AIOps/SOC* domain (I6). Exclusion criteria: sub-threshold venues (E1), non-agentic approaches (E2), out-of-domain studies (E3), duplicates (E4), and non-peer-reviewed material (E5). Operationally, the first three cycles expressed the venue cut through the Brazilian Qualis classification with SJR quartiles as auxiliary input; for international validity the authors later ratified the criterion directly on the Scopus percentile scale (a dated protocol decision), under which every historical eligibility decision was re-read: all previously caveated inclusions at Q1–Q3 stand without a venue caveat, and the three inclusions below the bar keep a mandatory caveat (Section IV-C). A dated protocol amendment in cycle 2 allowed four candidates outside the then-current strata to proceed to full-text assessment *with a*

mandatory caveat; the amendment is itself part of the audit trail and, notably, is the root cause of three of the eight residual inter-assistant divergences (Section IV-F).

Cycle 3 brought the review's first conference papers (26 of 37 candidates), for which the journal-oriented I3/I4 strata are undefined. Two further dated decisions were ratified before screening: the stratum of a conference paper is derived from the *Scopus CiteScore percentile of its proceedings series* mapped through the same quartile scale used for journals, with series lacking a CiteScore ineligible at screening; and the stratum amendment was renewed for the cycle, so sub-threshold candidates again proceeded with the mandatory caveat. Finally, five accepted candidates (Springer proceedings chapters) whose full texts could not be obtained through the institutional subscription after two documented attempts were removed under the PRISMA 2020 ''reports not retrieved'' category — an exclusion by unavailability, not by merit.

E. RATIFIED PROTOCOL RULES (R1–R3)

The inter-assistant agreement analysis (Section IV-F) traced every residual divergence among the LLM assistants to three recurring boundary questions. After adjudicating those cases, the authors ratified the following rules, which are consistent with — and now normatively bind — the published corpus decisions:

- **R1 (non-agentic secondary studies).** Secondary studies (surveys/SLRs) whose object is *not* Agentic AI are excluded from the primary corpus regardless of methodological quality; agentic-centered secondary studies are retained as *foundational-conditional* background (e.g., [21], [22], [23], [16], [2]). This ratifies the exclusion of the non-agentic RCA and log-anomaly reviews assessed in cycle 1.
- **R2 (normative placement of amendments).** Eligibility amendments bind only when written into the *normative Stage-1 text* of the assessment protocol, not into its input metadata. The cycle-2 stratum amendment is ratified under this rule: the four sub-stratum candidates keep their full assessments with a mandatory stratum caveat attached to any use of their evidence.
- **R3 (domain boundary and the human-factors cluster).** Out-of-domain studies are excluded from the primary corpus even when methodologically strong; the human-factors cluster [24], [25], [26], [27] is retained strictly as *complementary evidence for the cognitive-load outcome* (Table 5) and never feeds the agentic research questions (RQ1, RQ2, RQ4); a cluster study with no RQ adherence at all remains excluded.

Cycle 3 was conducted *prospectively* under R1–R3: the ratified rules (including all active amendments) were rendered into the normative Stage-1 text of every assessment prompt from a single canonical template. The outcome is reported in Section IV-F: the cycle produced *zero* inter-assistant divergences, which we read as an empirical verification of the rules — in particular of R2, since the assistant that had halted on sub-stratum candidates in cycle 2 assessed all of them in cycle 3 once the amendment was normative text.

Selection flow across the three update cycles.

Stage	Cycle 1	Cycle 2	Cycle 3
Identified	$\approx$51 $\rightarrow$ 21	49	37
Screened (eligibility)	20 (–1 dup.)	49	19
Excluded at screening	2 (E1)	27 (E2/E3)	19 (18 E1, 1 E3)
Reports not retrieved	—	—	5
Assessed in full text	18	22	13
Excluded after assessment	4	2	1
Included	**14**	**20**	**12**
Corpus (cumulative)	33	53	**65**

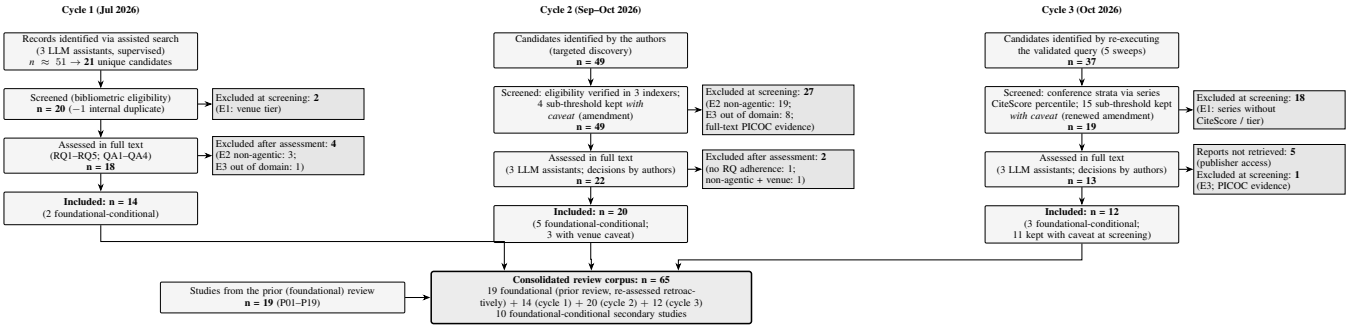
F. SELECTION FLOW (PRISMA)

Fig. 1 shows the PRISMA 2020 flow of the three update cycles over the foundational corpus, and Table 2 summarizes the stage counts side by side. From cycle 2 onward, thematic exclusion (E2/E3) was applied as early as the screening stage, because a full-text PICOC extraction of every retrievable candidate (one LLM reading agent per PDF, consolidated and audited by the authors) provided per-study evidence before assessment — each screening exclusion, decided by the authors, cites that evidence and a precedent from an earlier cycle.

The 65-study corpus comprises the 19 foundational studies, 14 cycle-1, 20 cycle-2, and 12 cycle-3 inclusions; ten of these are secondary ''foundational-conditional'' studies [21], [22], [23], [16], [28], [29], [2], [30], [31], [32] that migrate to background material if the protocol is restricted to primary studies (primary corpus: 55). Under the ratified percentile scale, three inclusions in the whole corpus (one Q4 series and two unindexed venues, all from cycle 3) carry the mandatory venue caveat; the other 62 sit at Q1–Q3 (53 Q1, 8 Q2, 1 Q3).

G. QUALITY ASSESSMENT AND EXTRACTION

Each assessed study received: (i) a verified bibliometric record (DOI, venue, year, and citation counts kept *per source* — OpenAlex, Crossref, Scopus — never summed, with ''not indexed'' distinguished from zero); (ii) an RQ-coverage score (RQ1–RQ5, each rated fully/partially/not answered = 1.0/0.5/0.0, max 5.0); (iii) a DARE-style quality score (QA1 clear objectives, QA2 replicable methodology, QA3 solid evidence base, QA4 coherent conclusions; max 4.0; bands High $\geq$3.0, Medium 1.5–2.5, Low $<$1.5); and (iv) an 11-field structured extraction sheet with per-claim evidence anchors (section/page). The 19 foundational studies were re-assessed retroactively under the same rubric — without re-opening their inherited inclusion — so that all 74 assessments are directly comparable.



PRISMA 2020 flow of the three update cycles over the foundational corpus (adapted for a review update). From cycle 2 onward, screening exclusions are anchored in a full-text PICOC extraction of every retrievable candidate; the venue amendments, the conference-series percentile rule, and the removal of the five unretrieved reports are dated decisions in the audit trail.

H. LLM-ASSISTED ASSESSMENT UNDER HUMAN DECISION

Every full-text assessment was drafted independently by three LLM-based research assistants (Claude, ChatGPT, Gemini) from the same written prompt and PDF, each without access to the others' outputs. The assistants' outputs are *inputs to*, not substitutes for, the authors' judgment: the master's student reviewed every drafted assessment, recorded the final include/exclude decision, and adjudicated every divergence, and the advisor reviewed and approved those decisions, the protocol amendments, and rules R1–R3. The agreement set comprises 74 assessments: the 19 foundational re-assessments, all 20 screened cycle-1 candidates (the two stratum-ineligible candidates also received full assessments, recorded as exclusions), the 22 cycle-2 assessments, and the 13 cycle-3 assessments. Agreement among the assistants was computed on their collapsed include/exclude recommendations with Cohen's κ per pair [11] and Fleiss' κ overall [12]; it serves as a consistency check that flags ambiguous cases for human adjudication, not as a measure of decision validity. Results are reported in Section IV-F.

Assistant versions. All runs used the same canonical prompt and the full-text PDF, under each interface's default settings (temperature is not user-configurable): Claude through the Claude Code interface with the PDF read as a file, ChatGPT and Gemini through their web applications with the PDF uploaded. The assistant versions changed across the review's timeline, as recorded in Table 3; Section VI discusses the implications. As a protocol improvement ratified after the third cycle, every assistant response from the next cycle onward must close with a mandatory footer recording the exact model version and the execution timestamp, so that version provenance is captured at generation time rather than reconstructed from logs.

LLM assistant versions per assessment set.

Assessment set	Claude	ChatGPT	Gemini
Cycle 1 (Jun. 2026)	Opus 4.8	GPT-5.4	Gemini 3.1 Pro
Cycle 2 (Sep. 2026)	Fable 5	GPT-5.6	Gemini 3.8 Flash
Foundational (Oct. 2026)	Fable 5	GPT-6.0	Gemini 3.8 Flash
Cycle 3 (Oct. 2026)	Fable 5	GPT-6.0	Gemini 3.8 Flash

A. CORPUS OVERVIEW AND TAXONOMY

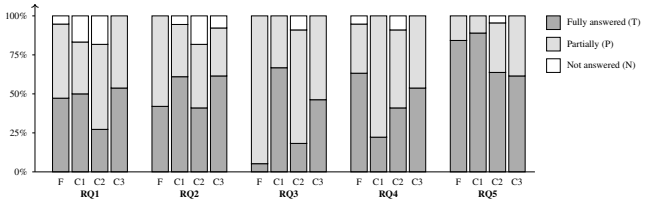
Table 4 organizes the 65 included studies by agentic paradigm. Four observations structure the corpus. First, the *foundational layer* is dominated by definitional surveys [3], [4], [15], [16] and governance/trust reviews [33], [34]; these answer RQ1/RQ4/RQ5 strongly but contribute no empirical benefit evidence. Second, the *first empirical core* is concentrated in cycle 1: multi-agent LLM systems for RCA and SOC [35], [17], [36], [37], closed-loop remediation [38], and a small-language-model agent for ICT operations [39] — the methodological high-water marks of the first two cycles (QA 3.5–4.0 of 4.0). Third, cycle 2 adds the copilot pattern proper — RAG-based security copilots [40], [41], [42] and SOC hyper-automation [43] — plus a *human-factors cluster* absent from earlier cycles [24], [25], [26], [27]. Fourth, cycle 3 adds a *time-instrumented cluster*: supervisor- and multi-agent LLM frameworks for cloud incident response and SRE that, for the first time in the corpus, measure nominal time metrics against executed baselines [44], [45], [46], [47], alongside an MCP-based security framework for agent pipelines [48], agentic SOC evaluation and SOC-as-a-service frameworks [49], [50], a consultative RAG responder [51], a non-LLM rules-plus-ML agent architecture [52], and three agentic-centered surveys [30], [31], [32] — with [52], [45], [46] under the venue caveat (bottom-quartile or unindexed venues).

B. RQ COVERAGE AND QUALITY

Fig. 2 shows per-RQ coverage across the four study sets. RQ coverage shifted between cycles in a telling way. In cycle 1 (18 assessed; mean RQ score 3.83, mean QA 3.50) the systematic gap was **RQ4** (ethics and governance: only 4/18 fully answered). In cycle 2 (22 assessed; mean RQ 3.16, mean QA 2.73; zero unconditional inclusions) the gap moved to **RQ3/QA3**: abundant conceptual contribution with scarce empirical benefit evidence. The retroactive assessment of the foundational studies (mean RQ 3.66, mean QA 2.53) shows the same profile — high RQ coverage by surveys, low QA

Taxonomy of the included corpus (n=65; representative studies).

Paradigm	Studies
LLM multi-agent (RCA/SOC/SRE)	[35], [17], [36], [37], [43], [44], [46]
Single-agent LLM copilot (+RAG)	[53], [40], [41], [42], [51]
Supervisor/event-driven agentic SOC	[49], [45]
Closed-loop remediation agents	[54], [38], [55], [47]
SLM/compact-model agents	[39]
MAS/RL, rules+ML, self-organization (non-LLM)	[56], [57], [52]
Conceptual agentic frameworks (AIOps/PdM/SOCaaS)	[58], [59], [50]
Agent-pipeline security (MCP)	[48]
AI-assisted SIEM / alert triage	[60]
Foundational surveys (agents, LLM-sec, AIOps)	[14], [3], [4], [18], [15], [13], [21], [22], [23], [16], [28], [29], [2], [30], [31], [32]
Human factors & instruments	[61], [24], [25], [26], [27]



Coverage of the research questions by set (share of assessed studies rating each RQ fully/partially/not answered; F = foundational, n=19; C1 = cycle 1, n=18; C2 = cycle 2, n=22; C3 = cycle 3, n=13). The empirical core of cycle 1 (12/18) and the time-instrumented cycle 3 (6/13) are the sets in which RQ3 (evidence of benefits) is most often fully answered; the foundational surveys answer RQ3 fully in just 1/19 cases, and cycle 2 in 4/22 — while RQ4 (ethics and governance) is the systematic gap of cycle 1 (4/18) and RQ5 (gaps and directions) is strong everywhere.

for lack of their own validation. Cycle 3 (13 assessed; mean RQ 3.85, mean QA 2.77) is the first set since the cycle-1 core in which **RQ3 is no longer the weak point** (6/13 fully answered), driven by the time-instrumented cluster of Section IV-C; its QA profile, however, remains middling, dominated by testbed and synthetic evidence. Across the first 61 assessments the pattern was stable — *the literature is conceptually rich and empirically thin precisely on the benefit claims that motivate it* — and cycle 3 is the first measurable bend in that curve, led by early-stage venues.

C. EVIDENCE ON MTTD AND MTTR

This is the review's central result, and the one the third cycle materially changes. Through the first two cycles (89 candidates screened, 61 assessed in full text), **no study measured MTTD or MTTR nominally against its own executed baseline**: the single nominal MTTR datapoint — the SOC hyper-automation agent of [43], ≈1–1.5 min per phase in a proof of concept compared against *literature values* for a commercial copilot (≈30–45 s) — had no baseline of its own and was unfavorable to the proposed agent, while secondary sources relayed dramatic but unreproduced gains (MTTR from 8 hours to 90 minutes; 98% auto-triage with sub-5-minute containment; a 277-day industry detect-and-contain baseline) [2].

Cycle 3 surfaces the corpus's **first cluster of primary studies with nominal time metrics against executed baselines**:

- **Production, human baseline.** The agentic SRE copilot of [44] (five LLM agents over a knowledge graph, with a guarded action layer) reports production p90 MTTD falling from 30 to 5 minutes (-83%) and p90 MTTR from 240 to 20 minutes (-91%; $\approx -63\%$ in the paper's aggregate figure) against experienced human SREs with dashboards and runbooks, at a reported LLM cost of $0.079 per incident.
- **Testbed, ablated baselines.** The event-driven SOC with a supervisor agent of [45] reports MTTR of 71.7 s vs. 258.4 s for a polling baseline (-72.3%) and 190.4 s for an LLM-assist (advisory-only) ablation (-62.3%), over 1,247 synthetic events drawn from MITRE ATT&CK scenarios. The multi-agent SRE framework of [46] reports MTTD of 8.5 vs. 47.2 minutes (-82%) against a consensus baseline of three senior SREs — MTTR itself is not measured.
- **Public dataset, internal baseline.** The adaptive counterfactual-validation framework of [47] reports mean time-to-containment of 8.9 vs. 14.3 minutes (-37.8%, Wilcoxon $p < 0.001$) against a SOAR baseline executed by the authors on the same 143 DARPA OpTC alerts; its external baselines are literature-derived and explicitly indicative only.

Three qualifications bound this cluster, and they are the reason the review's conclusion changes in degree rather than in kind. First, *venue tier*: under the review's percentile criterion, [47] (a Q2 journal) and [44] (a Q3 proceedings series) are eligible outright, while [45] (a bottom-quartile series) and [46] (a journal not indexed in Scopus) sit below the Q1–Q3 bar and carry the mandatory venue caveat — so half of the cluster, and in particular all of its *testbed* MTTR evidence, remains below the bar, and none of it yet appears in a top-quartile venue. Second, *setting*: only [44] is a production deployment, and it is a single, vendor-affiliated one; [45], [46] are testbeds and [49] (which reports MTTD/MTTR reductions of 44–80% for an agentic SOC against a traditional-SOC arm) runs both arms on synthetic telemetry with internally inconsistent tables, recorded in its assessment. Third, *assessment*: the three LLM assistants unanimously recommended all four studies for inclusion, and the authors confirmed the inclusions with the applicable caveats attached. Outside this cluster, proxies still dominate: a 74–84% alert-volume reduction in a national SOC [60], operator reaction time falling from ≈276 s to ≈107 s with decision support [27], report latency of 2.9–11.6 s [42], and answer quality without timing [41], [51]; detection studies report accuracy/F1/AUC only.

The structural mismatch diagnosed after cycle 2 — a field whose motivating claim is time-metric reduction but whose measurement practice is detection proxies — is therefore

beginning to close, from the early-stage end of the publication spectrum first. As of late 2026, the claim that agentic copilots reduce MTTD and MTTR is *supported by early primary evidence — half of it below this review's venue bar, none of it top-quartile, multi-site, or independently replicated*.

D. COGNITIVE LOAD: THE COMPLEMENTARY OUTCOME

The second gap concerns an outcome this review treats as complementary rather than incidental: operator cognitive load. The motivating observation was already present in the foundational review — the explosion of observability data (metrics, logs, traces) has raised the cognitive load of operations teams faster than tooling has relieved it [14], [13] — yet, within IT incident response, *no corpus study measures cognitive load directly*; alert fatigue appears only as an alert-volume proxy [60]. Direct measurement does exist in adjacent domains, and cycle 2 deliberately retained this human-factors cluster (under a dated protocol amendment): neuroergonomic foundations of mental workload and engagement [24]; a multimodal workload database with classification AUC up to 0.998 and NASA-TLX ground truth [26]; and a human experiment in which an AI decision-support system improved control-room operator performance [27]. The corpus also offers a validated usability instrument for conversational agents (BUS-15) [25]. Cycle 3 does not change this picture: its time-instrumented cluster measures system latencies, not operator state, and its one human-facing evaluation scores answer quality with LLM-as-judge rubrics rather than workload [51].

Table 5 consolidates these assets into a *measurement toolbox* for future copilot studies — in our view, the most reusable contribution this corpus makes to the cognitive-load question. The decisive observation is that the intersection *agentic copilot* × *IT incident response* × *measured cognitive load* is empty: every instrument in Table 5 has been validated, but never in the presence of an agentic copilot in IT operations. Together with Section IV-C, this defines the experimental agenda that motivates the authors' ongoing work: a copilot evaluated in the field with nominal MTTD/MTTR against its own baseline *and* the instruments of Table 5.

A cognitive-load measurement toolbox for agentic-copilot studies, assembled from the corpus.

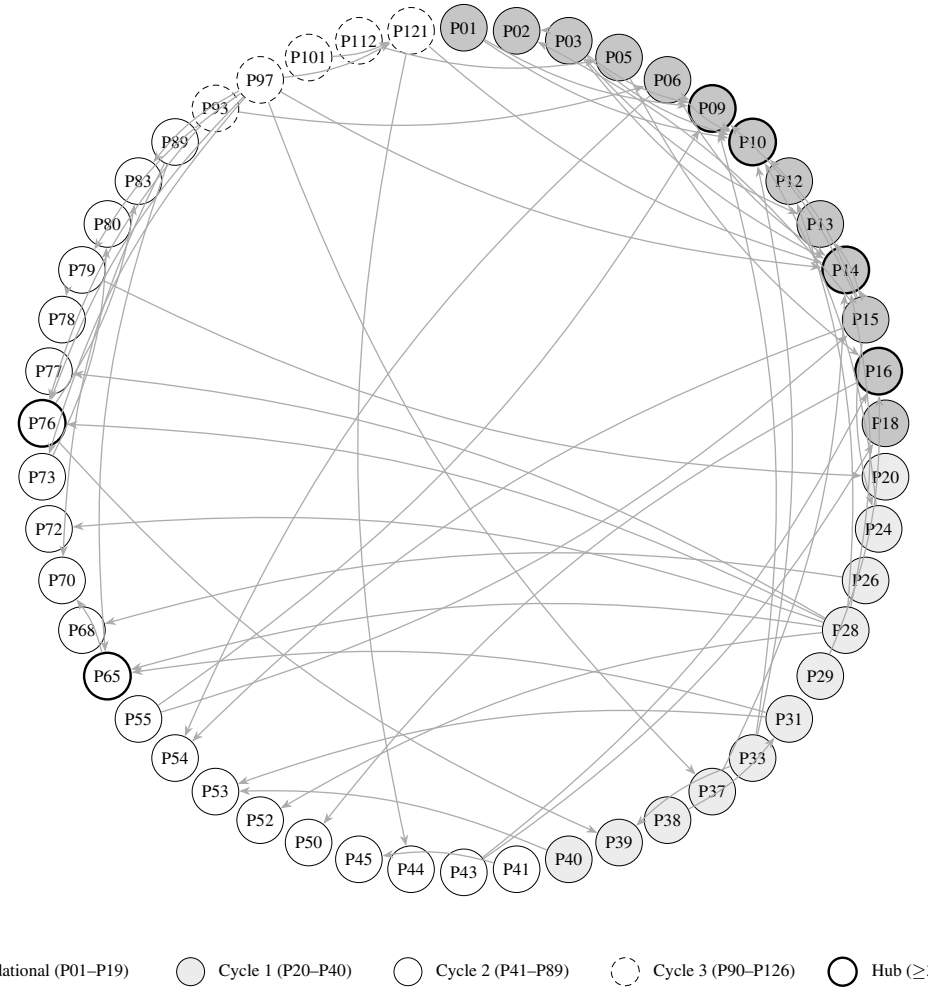
Instrument	Corpus evidence	Recommended use with a copilot
NASA-TLX (subjective workload)	Ground truth for multimodal workload classification [26]	Pre/post per incident shift, copilot on vs. off
Physiological multimodal (EEG, ECG, breathing)	Classification AUC up to 0.998 under activity [26]; neuroergonomic framework [24]	Continuous workload index during on-call duty; validate against NASA-TLX
Reaction / response time	Operator reaction time ≈276→107 s with AI decision support (Wilcoxon $p < 0.01$) [27]	Time-to-first-action per alert as the operator-level analogue of MTTD
Alert-volume exposure	74–84% global alert reduction in a national SOC [60]	Alerts per analyst-hour as the fatigue *exposure* covariate
Conversational usability (BUS-15)	Validated 15-item scale for AI-based conversational agents [25]	Perceived quality of the copilot interface; moderator of workload effects

E. CROSS-CITATION STRUCTURE OF THE CORPUS

Internal citation analysis over the full corpus (through cycle 2: triple-checked across OpenAlex `referenced_works`, Crossref reference lists, and Scopus `REF()` by title; cycle-3 additions: union of a full-text reference scan of each included PDF and OpenAlex-deposited references; 65 citing edges in total) shows a coherent field rather than a disconnected collection of papers. The hubs are the definitional surveys — [4] and [18] (7 internal citations each), [3] (6) — followed by the AIOps/MLOps survey [13] and the SOC hyper-automation study [43] (3 each; the log-anomaly survey [62], a screened candidate outside the corpus, also receives 3). The most connected citers are the multi-agent SOC system of [17] (7 outgoing edges, five of them to cycle-2 candidates) and the cycle-3 SOC-as-a-service framework of [50] (7 outgoing edges, all into earlier cycles or intra-cycle), and each cycle's studies cite back into the corpus ([2]→[54], [40]; [43]→[63], [41]; [49]→[64], [43]), indicating that the cycles sample one evolving conversation. Source-specific quirks are material to reproducibility: Scopus resolves references by *title*, not DOI; one open-access venue is not indexed there at all; and preprint DOIs can split citation identity across sources (handled here by alias resolution). Fig. 3 renders the full network. The cycle-3 inclusions add 13 edges and deepen the pattern: the SOC-as-a-service framework [50] cites seven corpus studies — among them [18], [60], [43], [2] and the intra-cycle survey [32] — tying the multi-agent SOC system [17] as the most connected citer; the agentic-SOC evaluation [49] cites [64], [43]; and the cycle-3 surveys cite back into the definitional layer ([31]→[14]; [32]→[18]), and the event-driven SOC [45] cites the intra-cycle survey [32]. Two consequences for the hub structure: the SOC hyper-automation study [43], cited by [17], [49], [50], becomes a hub, and [18] rises to 7 internal citations, level with [4].

F. AGREEMENT AMONG THE LLM ASSISTANTS

Table 6 reports agreement among the three LLM assistants over their 74 collapsed include/exclude recommendations; the final decision in every case was taken by the authors. Pairwise agreement is 91.9–93.2% (κ 0.58–0.62, with one pair reaching the *substantial* band); Fleiss' $\kappa = 0.60$; **66/74 recommendations are unanimous** (62 inclusions and 4 exclusions), including all 19 foundational re-assessments and the entire third cycle. The κ values are deflated by prevalence (90.1% of all verdicts are "include"), so the substantive reading is the nature of the eight divergences: *all eight are protocol-rule divergences, none is a content divergence, and all eight predate cycle 3* — in every case the three assistants'



Internal citation network of the corpus (65 edges, A→B = "A cites B"; edges through cycle 2 triple-checked across OpenAlex, Crossref, and Scopus; cycle-3 edges from full-text reference scans united with OpenAlex deposits). Definitional surveys [4], [18], [3] are the main hubs, now joined by the SOC hyper-automation study [43]; the multi-agent SOC system [17] and the cycle-3 SOC-as-a-service framework [50] are the most connected citers (7 outgoing edges each), and each new cycle cites back into the corpus — evidence of one evolving research conversation rather than disjoint samples. Node identifiers are mapped to references in the Appendix (Table 7). Nodes marked ‡ in Table 7 are screened candidates or excluded studies that take part in citation edges but are not part of the included corpus; included studies without internal edges are omitted.

scores agree and only the consequence rule differs. They fall into exactly three families: (A) the treatment of non-agentic secondary studies (2 cases); (B) whether the cycle-2 stratum amendment binds, which one assistant — applying the strict criterion — did not honor (3 cases); and (C) domain/stratum boundary calls (3 cases). The authors adjudicated all eight cases, and the three families were ratified as protocol rules R1–R3 (Section III-E) and cycle 3 was run prospectively under them: its 13 assessments came out **unanimous 13/13** ($\kappa = 1.00$ in every pair within the cycle), including the unanimous thematic exclusion of the one screening-flagged borderline study — an empirical verification that the rules, once normative, eliminate the divergence classes they name. Mean absolute score differences remain small (RQ 0.43–0.61; QA 0.31–0.61), with stable assistant profiles: one strict on protocol and quality, one lenient on scores (though its quality bands matched the intermediate assistant's exactly in cycle 3), one intermediate with the hardest thematic cut. Because

Agreement among the three LLM assistants on include/exclude recommendations (n=74; final decisions by the authors).

	A×B	A×C	B×C
Agreement	91.9%	93.2%	93.2%
Cohen's κ	0.62	0.58	0.58

Fleiss' κ (three assistants): **0.60**; unanimous recommendations: **66/74** (cycle 3 alone: 13/13 unanimous; $\kappa = 1.00$ per pair).

90.1% of the recommendations are "include", we also report prevalence-robust coefficients: Gwet's AC1 [65] of 0.91 (95% CI 0.85–0.97) and PABAK [66] of 0.86 (95% CI 0.76–0.95); interpretation bands follow Landis and Koch [67].

A. WHAT THE EVIDENCE SUPPORTS — AND WHAT IT DOES NOT

The corpus firmly supports three claims: (i) a consolidated conceptual model of Agentic AI for operations (autonomy levels, orchestration, guardrails, HITL) [3], [4], [15], [16], [31]; (ii) technical feasibility of agentic RCA, triage, and remediation at prototype level, with strong detection-quality results [39], [38], [35], [17], [36], [37], [46]; and (iii) a well-articulated risk and governance agenda [33], [68], [34], [18], [48]. The headline operational claim now has its first primary support — the cycle-3 cluster of Section IV-C — but that support is early, partly below the venue bar, and production-grade in exactly one vendor-affiliated deployment. Reviewers and practitioners should still treat vendor-scale numbers (e.g., 8h $\rightarrow$ 90min MTTR) as hypotheses under test: the cycle-3 measurements are the first tests, not yet the established effect.

B. WHY THE MEASUREMENT GAP PERSISTED — AND WHERE IT IS CLOSING

Three causes were visible through cycle 2: production access (most studies evaluate on public datasets or testbeds; industrial deployments are rare and rarely instrumented for time metrics); benchmark culture (detection communities optimize F1/AUC, which do not compose into end-to-end time outcomes); and the autonomy–accountability tension (studies that deploy agents constrain them to low-risk actions, making time gains modest and hard to attribute). That the only pre-cycle-3 nominal measurement [43] was *unfavorable* to the agent remains a useful counterweight to publication bias. Cycle 3 adds a fourth, sociological observation: the first time-instrumented studies are appearing in early-stage venues — regional conference series and recently launched journals — ahead of the top-quartile literature. A review that dropped every below-bar candidate without the caveat mechanism would have seen only half of the cluster; the dated-amendment design is what keeps the field's actual turning point visible while keeping the evidence grade explicit.

C. RESEARCH GAPS CONSOLIDATED ACROSS THE REVIEWS

The foundational review synthesized six gap axes for agentic IR; the three update cycles allow each to be re-examined against newer evidence. **(i) From recommendation to safe autonomous execution.** Remediation in LLM4AIOps was found to stop at recommendation and triage [14], [15]; cycles 1–2 confirmed the axis — closed-loop execution at prototype level [38], [55] and at proof-of-concept level in SOC hyperautomation [43] — and cycle 3 moves it furthest: guarded production execution with human gates [44], event-driven SOAR tool-calling [45], and counterfactual pre-execution validation with adaptive re-planning [47]. Verifiable, auditable execution beyond a single vendor deployment remains unreported. **(ii) Complex telemetry (traces).** The difficulty of feeding distributed traces to LLMs [14] persists: cycle-2 primary studies operate on logs, alerts, and flows [69], [60], [42]; trace-native agentic RCA remains open. **(iii) Memory integrity and contradictory knowledge.** Memory poisoning and conflicting sources were flagged as structural risks [33]; no update-cycle study evaluates agent-memory integrity empirically — the axis is untouched. **(iv) Governance, accountability, and adversarial risk.** The richest axis in volume: governance frameworks and threat taxonomies [33], [68], [18], [34], [28], [29] and HITL adoption factors [70], [61], yet still without quantitative evaluation of governance mechanisms under incident conditions. **(v) Causal reasoning vs. statistical correlation.** Agentic RCA advanced with graph- and Bayesian-structured approaches [55], [37], [69], partially addressing the axis, though evaluation remains on curated failure sets. **(vi) Resource efficiency and sustainability.** The SLM agent of [39] is still the corpus's only direct response (compact models, confidentiality, cost); the axis otherwise remains a statement of intent. Axes (i), (iii), and (vi) are therefore the least served by three years of literature — a concrete map for future primary studies, to which this review adds a seventh axis: *measured cognitive-load outcomes* (Section IV-D).

D. IMPLICATIONS FOR RESEARCH AND PRACTICE

For researchers, the agenda is concrete: replicate the cycle-3 cluster at this review's venue-quality bar — field studies that instrument MTTD and MTTR nominally, pre/post or A/B against the team's own baseline, multi-site rather than single-vendor [44], [45], [46], [47] — and pair time metrics with validated human-factors instruments (NASA-TLX [26], BUS-15 [25]) to test the cognitive-load "bonus" that no study has yet measured. For practitioners, the corpus supports piloting copilots on triage and report drafting (where latency and quality evidence exists [41], [42], [51]) under HITL governance [70], [61], now with early quantitative guidance on attainable time reductions and their cost [44], [45], while treating full-loop autonomous remediation as experimental [38], [43], [47] and securing the agent pipeline itself [48].

E. METHODOLOGICAL LESSONS FOR LLM-ASSISTED SLRS

Three findings generalize beyond this topic. First, three independently run LLM assistants under one written protocol, supervised by human reviewers who keep every decision, achieve high percentage agreement (91.9–93.2% pairwise) with moderate-to-substantial chance-corrected κ, and their residual disagreements expose *protocol ambiguities, not model randomness* — a cheap and effective mechanism for directing human attention to the cases that need adjudication. Second, protocol amendments must be placed in the normative body of the evaluation prompt, not in its input metadata: the one assistant that missed the cycle-2 stratum amendment read it as advisory, producing three of the eight divergences. Third, the fix is verifiable: once the ratified rules were rendered into the normative prompt text from a canonical template, the next full cycle (13 assessments, including sub-stratum candidates and a flagged borderline exclusion) ran *divergence-free* — protocol maturation in this design is

observable as a measured drop in disagreement, not merely asserted.

Selection. Assisted identification (LLM search in cycle 1, author-driven discovery in cycle 2, query re-execution in cycle 3) may bias candidate coverage; this is mitigated by dual-engine string validation (13/14 recall on OpenAlex and Scopus), mandatory snowballing (executed in cycle 3 with a negative result), and full audit trails, but a database-exhaustive census was not the goal of the update design. Five cycle-3 candidates were removed as PRISMA ''reports not retrieved'' after two documented access attempts — an unavailability exclusion whose effect on the synthesis is unknowable by construction. **Venue criteria.** The venue bar is the Scopus CiteScore percentile in quartiles (Q1–Q3 eligible), ratified by the authors for international validity; the first three cycles operationalized the same cut through the Brazilian Qualis classification with SJR as auxiliary input, and every dated amendment that retained below-bar studies is explicit in the audit trail. All conclusions hold with the three below-bar inclusions removed, with one quantified consequence: the cycle-3 nominal-metric cluster narrows from four studies to two ([44], [47]), losing its testbed MTTR evidence — which is precisely why the evidence grade is reported alongside the effect sizes, and why the one production deployment is flagged as vendor-affiliated and single-site. **Assistant validity.** The three assessment assistants are LLM systems; they run independently of each other but may share model-family biases (training data, sensitivity to the shared prompt) and may misread or misreport content. The main mitigation is the human-decision design: no screening, inclusion, exclusion, or adjudication was delegated to an LLM — all were taken by the authors (the master's student, reviewed by the advisor) — and per-claim evidence anchors (section/page) in every assessment allow each claim to be traced to its source during that review. The unanimity of the core corpus (89%), the protocol-only nature of the divergences, and a divergence-free third cycle under the ratified rules indicate consistency among the assistants, not by themselves correctness; the three protocol questions exposed by the divergences were adjudicated by the authors and ratified as rules R1–R3 (Section III-E). The assistant versions also changed across the review's timeline (Table 3: Claude Opus 4.8 to Fable 5, GPT-5.4 to GPT-6.0, Gemini 3.1 Pro to 3.8 Flash); agreement did not degrade across these updates — cycle 3, run entirely on the newer versions, was divergence-free — indicating that the observed consistency rests on the written protocol rather than on any specific model build. **Bibliometric sources.** Citation counts are kept per source and never summed; source quirks (title-based reference resolution, unindexed venues, preprint DOI splits, API service-level limits) are documented in the audit trail.

This SLR consolidated 65 studies (2020–2026) on Agentic AI as a copilot for incident response across three update cycles, through an LLM-assisted, human-decided protocol: three supervised LLM assistants drafted each full-text assessment with high mutual agreement (66 of 74 unanimous; the third cycle divergence-free), and every decision was taken and reviewed by the authors. The field has matured conceptually — shared definitions, reference architectures, governance agendas, and a coherent citation structure — and demonstrates prototype-level feasibility of agentic detection, RCA, and remediation. Its motivating claim is now beginning to be measured: a first cluster of four primary studies reports nominal reductions against executed baselines — MTTD by 82–83%, MTTR by 63–72%, and time-to-containment by 38%, in settings ranging from one production deployment to testbeds and a public dataset — half of it below the review's venue bar, with production evidence from a single vendor-affiliated deployment, and still without any measurement of operator cognitive load in the presence of an agentic copilot. A useful next step is a multi-site field evaluation involving operations teams beyond a single vendor. It should measure MTTD and MTTR against an explicit baseline, include validated workload instruments, and report methods and results in enough detail for independent appraisal.

Throughout this review — in particular in the PRISMA flow (Fig. 1) and in the citation network (Fig. 3) — studies are referred to by their stable protocol identifiers (P01–P126), which are also the keys of all machine-readable artifacts in the audit trail. Table 7 maps every identifier that appears in this article to its bibliography entry. Unmarked identifiers are included corpus studies; † marks studies assessed in full text and excluded; ‡ marks screened candidates that appear only as nodes of the citation network.

The authors thank the Applied Computing Graduate Program (PPGCA) at Unisinos. LLM-based assistants (Claude, ChatGPT, Gemini) were used as described in Section III; all decisions, analyses, and conclusions were reviewed by the authors, who take full responsibility for the content of this article. Review artifacts (protocol, prompts, assessments, extraction sheets, and machine-readable tables) are maintained in a version-controlled repository and are available from the authors upon reasonable request.

Competing interests. The first author is employed in IT operations leadership at a communications-platform company; the company had no role in the design, conduct, or reporting of this review. The authors declare no other competing interests.

[1] IBM Security and Ponemon Institute, ''Cost of a Data Breach Report 2023,'' IBM Corporation, Tech. Rep., 2023.

Study identifiers (Pxx) and their references.

ID	Ref.	ID	Ref.	ID	Ref.	ID	Ref.
P01	[33]	P22	[38]	P45‡	[71]	P82	[59]
P02	[72]	P23	[73]	P50‡	[74]	P83	[41]
P03	[75]	P24	[21]	P52	[23]	P85†	[76]
P04	[77]	P25	[56]	P53	[69]	P86†	[78]
P05	[14]	P26†	[79]	P54	[16]	P87	[42]
P06	[64]	P27	[35]	P55	[80]	P88	[57]
P07	[81]	P28	[17]	P62	[24]	P89‡	[82]
P08	[70]	P29†	[83]	P63	[25]	P91	[52]
P09	[3]	P30†	[84]	P65‡	[62]	P93	[49]
P10	[4]	P31	[36]	P68‡	[85]	P94	[48]
P11	[86]	P32	[55]	P69	[26]	P97	[50]
P12	[68]	P33	[22]	P70‡	[87]	P100	[44]
P13	[88]	P34	[53]	P72	[28]	P101	[45]
P14	[18]	P35	[37]	P73	[60]	P102	[46]
P15	[15]	P37	[61]	P74	[89]	P110	[30]
P16	[13]	P38†	[90]	P76	[43]	P112	[31]
P17	[34]	P39†	[63]	P77	[29]	P119	[51]
P18	[91]	P40†	[92]	P78	[40]	P121	[32]
P19	[93]	P41‡	[94]	P79	[2]	P123	[47]
P20	[54]	P43	[58]	P80‡	[95]		
P21	[39]	P44‡	[96]	P81	[27]		

[2] S. Srinivas, B. Kirk, J. Zendejas, M. Espino, M. Boskovich, A. Bari, K. Dajani, and N. Alzahrani, "AI-Augmented SOC: A Survey of LLMs and Agents for Security Automation," *Journal of Cybersecurity and Privacy*, 2025.

[3] R. Sapkota, K. I. Roumeliotis, and M. Karkee, "AI Agents vs. Agentic AI: A Conceptual taxonomy, applications and challenges," *Information Fusion*, 2025.

[4] D. B. Acharya, K. Kuppan, and B. Divya, "Agentic AI: Autonomous Intelligence for Complex Goals — A Comprehensive Survey," *IEEE Access*, 2025.

[5] B. Kitchenham and S. Charters, "Guidelines for Performing Systematic Literature Reviews in Software Engineering," Keele University and Durham University, Tech. Rep. EBSE-2007-01, 2007.

[6] M. J. Page *et al.*, "The PRISMA 2020 statement: an updated guideline for reporting systematic reviews," *BMJ*, vol. 372, p. n71, 2021.

[7] M. Petticrew and H. Roberts, *Systematic Reviews in the Social Sciences: A Practical Guide*. Blackwell Publishing, 2006.

[8] Centre for Reviews and Dissemination, "Database of Abstracts of Reviews of Effects (DARE)," University of York, 2002.

[9] T. Dybå and T. Dingsøyr, "Empirical studies of agile software development: A systematic review," *Information and Software Technology*, vol. 50, no. 9–10, pp. 833–859, 2008.

[10] E. Mendes, C. Wohlin, K. Felizardo, and M. Kalinowski, "When to update systematic literature reviews in software engineering," *Journal of Systems and Software*, vol. 167, p. 110607, 2020.

[11] J. Cohen, "A coefficient of agreement for nominal scales," *Educational and Psychological Measurement*, vol. 20, no. 1, pp. 37–46, 1960.

[12] J. L. Fleiss, "Measuring nominal scale agreement among many raters," *Psychological Bulletin*, vol. 76, no. 5, pp. 378–382, 1971.

[13] J. D. de Arcaya, A. I. Torre-Bastida, G. Zárate, R. Miñón, and A. Almeida, "A Joint Study of the Challenges, Opportunities, and Roadmap of MLOps and AIOps," *ACM Computing Surveys*, 2023.

[14] L. Zhang, T. Jia, M. Jia, Y. Wu, A. Liu, Y. Yang, Z. Wu, X. Hu, P. Yu, and Y. Li, "A Survey of AIOps in the Era of Large Language Models," *ACM Computing Surveys*, 2025.

[15] A. Bandi, B. Kongari, R. Naguru, S. Pasnoor, and S. V. Vilipala, "The Rise of Agentic AI: A Review of Definitions, Frameworks, Architectures, Applications, Evaluation Metrics, and Challenges," *Future Internet*, 2025.

[16] M. A. Ferrag, N. Tihanyi, and M. Debbah, "From LLM Reasoning to Autonomous AI Agents: A Comprehensive Review," *IEEE Access*, 2026.

[17] Y. Hmimou, M. Tabaa, A. Khiat, and Z. Hidila, "A Multi-Agent System for Cybersecurity Threat Detection and Correlation Using Large Language Models," *IEEE Access*, 2025.

[18] N. Kshetri, "Transforming cybersecurity with agentic AI to combat emerging cyber threats," *Telecommunications Policy*, 2025.

[19] M. L. Rethlefsen, S. Kirtley, S. Waffenschmidt, A. P. Ayala, D. Moher, M. J. Page, and J. B. Koffel, "PRISMA-S: an extension to the PRISMA Statement for Reporting Literature Searches in Systematic Reviews," *Systematic Reviews*, vol. 10, no. 1, p. 39, 2021.

[20] C. Wohlin, "Guidelines for snowballing in systematic literature studies and a replication in software engineering," in *Proc. 18th Int. Conf. Evaluation and Assessment in Software Engineering (EASE)*, 2014, pp. 1–10.

[21] F. Piccialli, D. Chiaro, S. Sarwar, D. Cerciello, P. Qi, and V. Mele, "AgentAI: A comprehensive survey on autonomous agents in distributed AI for industry 4.0," *Expert Systems with Applications*, 2025.

[22] I. ADABARA, B. O. Sadiq, A. N. Shuaibu, Y. I. Danjuma, and M. Venkateswarlu, "A Review of Agentic AI in Cybersecurity: Cognitive Autonomy, Ethical Governance, and Quantum-Resilient Defense," *F1000Research*, 2025.

[23] I. Hasanov, S. Virtanen, A. Hakkala, and J. Isoaho, "Application of Large Language Models in Cybersecurity: A Systematic Literature Review," *IEEE Access*, 2024.

[24] F. Dehais, A. Lafont, R. N. Roy, and S. Fairclough, "A Neuroergonomics Approach to Mental Workload, Engagement and Human Performance," *Frontiers in Neuroscience*, 2020.

[25] S. Borsci, A. Malizia, M. Schmettow, F. van der Velde, G. Tariverdiyeva, D. Balaji, and A. Chamberlain, "The Chatbot Usability Scale: the Design and Pilot of a Usability Scale for Interaction with AI-Based Conversational Agents," *Personal and Ubiquitous Computing*, 2021.

[26] I. Albuquerque, A. Tiwari, M. Parent, R. Cassani, J.-F. Gagnon, D. Lafond, S. Tremblay, and T. H. Falk, "WAUC: A Multi-Modal Database for Mental Workload Assessment Under Physical Activity," *Frontiers in Neuroscience*, 2020.

[27] J. Mietkiewicz, A. N. Abbas, C. W. Amazu, G. Baldissone, A. Madsen, M. Demichela, and M. C. Leva, "Enhancing Control Room Operator Decision Making," *Processes*, 2024.

[28] W. Kasri, Y. Himeur, H. A. Alkhazaleh, S. Tarapiah, S. Atalla, W. Mansoor, and H. Al-Ahmad, "From Vulnerability to Defense: The Role of Large Language Models in Enhancing Cybersecurity," *Computation*, 2025.

[29] M. Uddin, M. Irshad, I. A. Kandhro, F. Alanazi, F. Ahmed, M. Maaz, S. Hussain, and S. S. Ullah, "Generative AI revolution in cybersecurity: a comprehensive review of threat intelligence and operations," *Artificial Intelligence Review*, 2025.

[30] O. Ait Ami, A. Boumahdi, and M. Azmi, "DevAIOps: A Systematic Literature Review on Integrating Explainable AI Agents into DevOps Pipelines," *Procedia Computer Science*, 2026.

[31] M. Bilal, J. Crowcroft, R. Wang, X. Xu, and S. Dustdar, "Large Language

Models for Agentic NetOps and AIOps: Architectures, Evaluation, and Assurance," *Computer Science Review*, 2026.
[32] A. Habibzadeh, F. Feyzi, and R. Ebrahimi Atani, "Large Language Models for Security Operations Centers: A Comprehensive Survey," *Journal of Electrical and Computer Engineering*, 2026.
[33] I. ADABARA, B. O. Sadiq, A. N. Shuaibu, Y. I. Danjuma, and M. Venkateswarlu, "Trustworthy agentic AI systems: a cross-layer review of architectures, threat models, and governance strategies for real-world deployment," *F1000Research*, 2025.
[34] V. Chamola, V. Hassija, A. R. Sulthana, D. Ghosh, D. Dhingra, and B. Sikdar, "A Review of Trustworthy and Explainable Artificial Intelligence (XAI)," *IEEE Access*, 2023.
[35] F. Fu, H. Ding, Y. Qin, J. Yu, and D. Xu, "Leveraging multi-agent framework for root cause analysis," *Complex and Intelligent Systems*, 2026.
[36] X. Ji, L. Zhang, W. Zhang, F. Peng, Y. Mao, X. Liao, and K. Zhang, "LEMAD: LLM-Empowered Multi-Agent System for Anomaly Detection in Power Grid Services," *Electronics*, 2025.
[37] H. Zou, Y. Zhao, X. Chen, L. Wang, J. Yu, and L. Yuan, "Graph-Augmented Multi-Agent Robust Root Cause Analysis in AIOps," *Computers, Materials and Continua*, 2026.
[38] V. Avgerinos, K. Ramantas, L. Alonso, and C. Verikoukis, "ARM: Autonomous Remediation and Management With LLM Agents for Intent-Driven Control," *IEEE IoT Journal*, 2026.
[39] N. Fukuda, H. Nozue, and H. Oishi, "Small Language Model Agent for the Operations of Continuously Updating ICT Systems," *IEEE Access*, 2025.
[40] F. Y. Loumachi, M. C. Ghanem, and M. A. Ferrag, "Advancing Cyber Incident Timeline Analysis Through Retrieval-Augmented Generation and Large Language Models," *Computers*, 2025.
[41] I. Ismail, R. Kurnia, F. Widyatama, I. M. Wibawa, Z. A. Brata, Ukasyah, G. A. Nelistiani, and H. Kim, "Enhancing Security Operations Center: Wazuh Security Event Response with Retrieval-Augmented-Generation-Driven Copilot," *Sensors*, 2025.
[42] C.-H. Chou, A. Sudheer, and Y. Park, "An LLM-Based Agentic Network Traffic Incident-Report Approach Towards Explainable-AI Network Defense," *Journal of Sensor and Actuator Networks*, 2026.
[43] I. Ismail, R. Kurnia, Z. A. Brata, G. A. Nelistiani, S. Heo, H. Kim, H. Kim, H. Kim, and H. Kim, "Toward Robust Security Orchestration and Automated Response in Security Operations Centers with a Hyper-Automation Approach Using Agentic Artificial Intelligence," *Information*, 2025.
[44] A. Pratinav and A. Kataria, "AIDR-Cloud: An Agentic AI-Driven Incident Response Framework for Cloud Failures," in *Proc. IEEE SoutheastCon*, 2026.
[45] E. Özgözler, A. Varol, and İ. Tanrıverdi, "Event-Driven Agentic SOC (ED-ASOC): Supervisor-Based LLM Framework for Dynamic Incident Response," in *Proc. Int. Symp. Digital Forensics and Security (ISDFS)*, 2026.
[46] V. Kataria, "Intelligent Site Reliability Engineering: A Multi-Agent LLM Framework for Automated Incident Detection," *International Journal of Intelligent Engineering and Systems*, 2025.
[47] S. P. Kizhakel and S. D. Pande, "An Adaptive Agentic AI Framework for Counterfactual Validation and Adversarial Learning in Incident Response," *Discover Artificial Intelligence*, 2026.
[48] R. Tavares de Pina Simões, X. Larriva-Novo, C. Sánchez-Zas, V. A. Villagrá, and A. I. Marín López, "Design of a Security Framework for Multi-Agent Systems Based on Model Context Protocol," *Applied Sciences*, 2026.
[49] M. K. Y. A. Wahed and N. M. Aljawarneh, "From Manual to Agentic Security Operations: A Multi-Dimensional Evaluation of AI-Driven SOC Architectures," *Journal of Information & Knowledge Management*, 2026.
[50] A. Y. Alhumaidi, F. A. Al-Qadda, A. Alsumayt, and M. Alshammari, "Agentic AI-Driven SOC-as-a-Service for Optimizing Incident Response in Cloud Environments: A Conceptual Framework," *Network*, 2026.
[51] A. Tellache, A. Amara Korba, A. Mokhtari, H. F. Moldovan, and Y. M. Ghamri-Doudane, "Advancing Autonomous Incident Response: Leveraging LLMs and Cyber Threat Intelligence," in *Proc. IEEE Global Communications Conference (GLOBECOM)*, 2025.
[52] M. Stefanov, K. Stefanov, L. N. Kandel, S. R. Crouse, and B. Jekov, "Autonomous Agentic AI Architectures for Optimizing Security Operations Centers (SOC) KPIs: Methodology, Impact on Detection, Response, and Recovery," *Land Forces Academy Review*, 2025.
[53] G. Jones, D. Kasimatis, N. Pitropakis, R. Macfarlane, and W. J. Buchanan, "Analysing the role of LLMs in cybersecurity incident management," *International Journal of Information Security*, 2025.
[54] D. Toprani and V. K. Madisetti, "LLM Agentic Workflow for Automated Vulnerability Detection and Remediation in IaC," *IEEE Access*, 2025.
[55] W. Zhang, Z. Yang, F. Peng, L. Zhang, Y. Chen, and R. Chen, "GALR: Graph-Based Root Cause Localization and LLM-Assisted Recovery for Microservices," *Electronics*, 2026.
[56] A. Agrawal, M. Nadeem, A. A. Nuaim, and A. A. Nuaim, "Artificial intelligence driven multi agent framework for adaptive cyber attack simulation and automated incident response in cyber range environments," *Scientific Reports*, 2026.
[57] A. Rodríguez, J. Gómez, and A. Diaconescu, "A Decentralised Self-Healing Approach for Network Topology Maintenance," *Autonomous Agents and Multi-Agent Systems*, 2020.
[58] R. D. Zota, C. Bărbulescu, and R. Constantinescu, "A Practical Approach to Defining a Framework for Developing an Agentic AIOps System," *Electronics*, 2025.
[59] W. Jiang and F. Hu, "Artificial Intelligence Agent-Enabled Predictive Maintenance: Conceptual Proposal and Basic Framework," *Computers*, 2025.
[60] T. Ban, T. Takahashi, S. Ndichu, and D. Inoue, "Breaking Alert Fatigue: AI-Assisted SIEM Framework for Effective Incident Response," *Applied Sciences*, 2023.
[61] O. I. Falowo and J. B. Abdo, "Empirical Study on Automation, AI Trust, and Framework Readiness in Cybersecurity Incident Response," *Algorithms*, 2026.
[62] M. Landauer, S. Onder, F. Skopik, and M. Wurzenberger, "Deep learning for anomaly detection in log data: A survey," *Machine Learning with Applications*, 2023.
[63] S. Oesch, J. Hutchins, P. Austria, and A. Chaulagain, "Agentic AI and the Cyber Arms Race," *Computer*, 2025.
[64] A. K. Pati, "Agentic AI: A Comprehensive Survey of Technologies, Applications, and Societal Implications," *IEEE Access*, 2025.
[65] K. L. Gwet, "Computing inter-rater reliability and its variance in the presence of high agreement," *British Journal of Mathematical and Statistical Psychology*, vol. 61, no. 1, pp. 29–48, 2008.
[66] T. Byrt, J. Bishop, and J. B. Carlin, "Bias, prevalence and kappa," *Journal of Clinical Epidemiology*, vol. 46, no. 5, pp. 423–429, 1993.
[67] J. R. Landis and G. G. Koch, "The measurement of observer agreement for categorical data," *Biometrics*, vol. 33, no. 1, pp. 159–174, 1977.
[68] M. A. Lopez, W. T. Lunardi, G. Lawton, and S. Thakkar, "Enhancing Autonomous System Security and Resilience With Generative AI: A Comprehensive Survey," *IEEE Access*, 2024.
[69] D. F. Pedroso, L. Almeida, L. E. G. Pulcinelli, W. A. A. Aisawa, I. Dutra, and S. M. Bruschi, "Anomaly Detection and Root Cause Analysis in Cloud-Native Environments Using Large Language Models and Bayesian Networks," *IEEE Access*, 2025.
[70] S. Kumar, S. Datta, V. Singh, D. Datta, S. K. Singh, and R. Sharma, "Applications, Challenges, and Future Directions of Human-in-the-Loop Learning," *IEEE Access*, 2024.
[71] K. Choi, J. Yi, C. Park, and S. Yoon, "Deep Learning for Anomaly Detection in Time-Series Data: Review, Analysis, and Guidelines," *IEEE Access*, 2021.
[72] S. Hosseini and H. Seilani, "The role of agentic AI in shaping a smart future: A systematic review," *Array*, 2025.
[73] X. Zhang, Q. Wang, M. Li, Y. Yuan, M. Xiao, F. Zhuang, and D. Yu, "TAMO:Fine-Grained Root Cause Analysis via Tool-Assisted LLM Agent With Multi-Modality Observation Data in Cloud-Native Systems," *IEEE Transactions on Services Computing*, 2025.
[74] M. Usman, S. Ferlin, A. Brunström, and J. Taheri, "A Survey on Observability of Distributed Edge and Container-Based Microservices," *IEEE Access*, 2022.
[75] S. N. Brohi, Q. ul-ain Mastoi, N. Z. Jhanjhi, and T. R. Pillai, "A Research Landscape of Agentic AI and Large Language Models: Applications, Challenges and Future Directions," *Algorithms*, 2025.
[76] F. C. Moreno, J. González, J. S. Muro, and R. López-Alonso, "Thermographic and cognitive assessment of fatigue in vessel traffic service operators: a socio-technical approach," *Cognition, Technology and Work*, 2025.
[77] J. He, C. Treude, and D. Lo, "LLM-Based Multi-Agent Systems for Software Engineering: Literature Review, Vision, and the Road Ahead," *ACM Transactions on Software Engineering and Methodology*, 2025.
[78] A. Poghosyan, A. Harutyunyan, N. Grigoryan, and C. C. C. Pang, "Distributed Tracing for Troubleshooting of Native Cloud Applications via Rule-Induction Systems," *Journal of Universal Computer Science*, 2023.

[79] Z. Li, J. Yu, Z. Huang, and Y. Huang, "Surveying Root Cause Analysis Techniques: A Comprehensive Review of Aspects for Multi-Service Applications," *IEEE Transactions on Services Computing*, 2025.
[80] S. Cruzes, "Telemetry and Agentic AI: Foundations for Optical Network Automation," *IEEE Access*, 2025.
[81] J. Chen, D. Zhang, Z. Qu, and C. Wang, "Artificial Empathy: A New Perspective for Analyzing and Designing Multi-Agent Systems," *IEEE Access*, 2020.
[82] M. Nasirzadeh, J. Tahmoresnezhad, and P. Rashidi-Khazaee, "A Unified Framework for Detecting Point and Collective Anomalies in Operating System Logs via Contrastive Learning," *Scientific Reports*, 2025.
[83] M. D. la Cruz Cabello, T. P. Sales, and M. Machado, "AIOps for log anomaly detection in the era of LLMs: A systematic literature review," *Intelligent Systems with Applications*, 2025.
[84] H. Li and Y. Wang, "Reliability of LLM Inference Engines from a Static Perspective: RCA and Repair Suggestion," *Big Data and Cognitive Computing*, 2026.
[85] Q. Ma, H. Li, and A. Thorstenson, "A Big Data-Driven Root Cause Analysis System: Application of Machine Learning in Quality Problem Solving," *Computers & Industrial Engineering*, 2021.
[86] A. Siddiqua, S. Liu, A. S. Nipu, A. M. Harris, and Y. Liu, "Co-Evolving Multi-Agent Transfer Reinforcement Learning via Scenario Independent Representation," *IEEE Access*, 2024.
[87] H. Studiawan, F. Sohel, and C. Payne, "Anomaly Detection in Operating System Logs with Deep Learning-Based Sentiment Analysis," *IEEE Transactions on Dependable and Secure Computing*, 2020.
[88] L. N. Mishra and B. Senapati, "Retail Resilience Engine: An Agentic AI Framework for Building Reliable Retail Systems With Test-Driven Development Approach," *IEEE Access*, 2025.
[89] S. Pahune and Z. Akhtar, "Transitioning from MLOps to LLMOps: Navigating the Unique Challenges of Large Language Models," *Information*, 2025.
[90] I. Radeva, I. Popchev, L. Doukovska, and M. Dimitrova, "Multi-Agent Coordination Strategies vs. Retrieval-Augmented Generation in LLMs: A Comparative Evaluation," *Electronics*, 2025.
[91] R. Eramo, B. Said, M. Oriol, H. Brunelière, and S. Morales, "An architecture for model-based and intelligent automation in DevOps," *Journal of Systems and Software*, 2024.
[92] J. Hong, N. Tu, and J.-K. Hong, "A Comprehensive Survey on LLM-Based Network Management and Operations," *International Journal of Network Management*, 2025.
[93] A. Tour, A. Polyvyanyy, and A. Kalenkova, "Agent System Mining: Vision, Benefits, and Challenges," *IEEE Access*, 2021.
[94] Z. Zamanzadeh Darban, G. I. Webb, S. Pan, C. C. Aggarwal, and M. Salehi, "Deep Learning for Time Series Anomaly Detection: A Survey," *ACM Computing Surveys*, 2024.
[95] S. Ndichu, T. Ban, T. Takahashi, and D. Inoue, "AI-Assisted Security Alert Data Analysis with Imbalanced Learning Methods," *Applied Sciences*, 2023.
[96] M. A. Ferrag, O. Friha, D. Hamouda, L. Maglaras, and H. Janicke, "Edge-IIoTset: A New Comprehensive Realistic Cyber Security Dataset of IoT and IIoT Applications for Centralized and Federated Learning," *IEEE Access*, 2022.

VALDOMIRO DE OLIVEIRA SOUZA JR. received the B.S. degree in information technology management from the Universidade do Vale do Rio dos Sinos (Unisinos), São Leopoldo, Brazil, in 2025, where he is currently pursuing the M.Sc. degree in applied computing with the Applied Computing Graduate Program (PPGCA).

From 1997 to 2012, he held infrastructure and operations positions in academic and industry organizations, including Pure Bros, a mobile value-added services integrator acquired by Zenvia in 2013. Since 2013, he has been with Zenvia, São Paulo, Brazil, where he has been Head of Information Technology since 2025. He leads the product, engineering, infrastructure, site reliability engineering (SRE), information security, observability, and FinOps teams of the company's communications platform as a service (CPaaS) business unit, whose messaging and voice platforms sustain peak loads above one million transactions per minute. He built Zenvia's SRE organization, which grew to 35 engineers across nine squads to support software as a service (SaaS), CPaaS, and corporate systems, led the improvements that supported the company's ISO/IEC 27001 certification in 2022, and was among the leaders of the technical integration of ten acquired companies. His research interests include agentic AI, AI-assisted incident response, the reduction of mean time to detect (MTTD) and mean time to recover (MTTR), and cognitive load in operations teams.

RAFAEL KUNST received the M.Sc. and Ph.D. degrees in computer science from the Federal University of Rio Grande do Sul (UFRGS), Porto Alegre, Brazil.

He is currently a Professor and a Researcher with the Applied Computing Graduate Program (PPGCA), Universidade do Vale do Rio dos Sinos (Unisinos), São Leopoldo, Brazil, where he currently serves as the Program Coordinator and is a member of the Software Innovation Laboratory (SOFTWARELAB). In 2020, he was a Visiting Professor with the IBM Watson IoT Center, Munich, Germany. He also collaborates with the Economics Graduate Program at Unisinos on data science topics and acts as an ad hoc Consultant for the Brazilian Ministry of Education. His research interests include next-generation mobile communications (5G and 6G), Industry 4.0, smart cities, the Internet of Things, and the application of big data analytics and machine learning to telecommunications.